MODULAR CURVES AND MAZUR–WILES THEORY

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ABSTRACT. Following Lang [2], this project develops the arithmetic of modular curves toward the Mazur–Wiles theorem [3] and the Iwasawa Main Conjecture for $\mathbb{Q}$. Starting from the action of $\mathrm{SL}_2(\mathbb{Z})$ on the upper half-plane and the j-invariant, we introduce the congruence subgroups $\Gamma(N)$, $\Gamma_1(N)$, $\Gamma_0(N)$ and their modular curves

$$Y(N),\ Y_1(N),\ Y_0(N), \qquad X(N),\ X_1(N),\ X_0(N),$$

which parametrize elliptic curves with level structure. At prime level p, the cuspidal divisor class group $\mathscr{C}_1^S(p)$ on the Jacobian $J_1(p)$ is identified by Kubert–Lang [7] with the co-Stickelberger module built from $\mathbf{B}_2(X) = X^2 - X + \frac{1}{6}$, yielding a class number formula in generalized Bernoulli numbers $B_{2,\chi}$. The Hecke algebra $\mathbf{T} \subset \mathrm{End}(J_1(p))$ acts on cuspidal torsion, and Wiles [4] and Mazur–Wiles [3] show that Hecke-annihilated torsion points generate unramified abelian extensions of $\mathbb{Q}(\mu_{p^n})$, realizing the Iwasawa Main Conjecture and recovering Herbrand–Ribet [5] as its base case.

CONTENTS

1. PRELIMINARIES

The theory we develop draws on several distinct mathematical traditions. On one side sit the complex-analytic objects — the upper half-plane, its discrete symmetry group, the j-invariant, and elliptic curves over $\mathbb{C}$ — which provide the geometric setting for modular curves. On the other sit the algebraic objects of number theory — Picard groups, Jacobian varieties, characters, and Bernoulli numbers — which encode the arithmetic content of cuspidal divisors. The central insight of Lang [2] is that these two worlds are the same world viewed from different angles: cyclotomic units correspond exactly to cuspidal divisors on modular curves. This section records what is needed from both traditions, following [14, 2].

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1.1. **The upper half-plane and the modular group.**

Definition 1.1.1 (Upper half-plane)**.** The *upper half-plane* and the *modular group* are

$$\mathscr{H} = \{\tau = x+iy \in \mathbb{C} \mid y>0\}, \qquad \Gamma(1) = \mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \,\middle|\, a,b,c,d \in \mathbb{Z},\ ad-bc=1 \right\}.$$

Definition 1.1.2 (Möbius transformation)**.** The *fractional linear action (Möbius transformation)* of $\Gamma(1)$ on $\mathscr{H}$ is

$$\gamma \cdot \tau = \frac{a\tau+b}{c\tau+d}, \qquad \gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

This is a left group action: $(\gamma_1\gamma_2)\cdot\tau = \gamma_1\cdot(\gamma_2\cdot\tau)$. The center $\{\pm I\}$ acts trivially, so the action descends to a faithful action of $\mathrm{PSL}_2(\mathbb{Z}) = \Gamma(1)/\{\pm I\}$.

Proposition 1.1.3 (Projective special linear group)**.** *The* projective special linear group $\mathrm{PSL}_2(\mathbb{Z})$ *has the presentation*

$$\mathrm{PSL}_2(\mathbb{Z}) \cong \langle S,T \mid S^2 = (ST)^3 = 1\rangle \cong \mathbb{Z}/2 * \mathbb{Z}/3,$$

stabilizers $\langle S\rangle, \langle T\rangle$ *allow* $S = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ *acts as* $\tau \mapsto -1/\tau$ *and* $T = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ *acts as* $\tau \mapsto \tau+1$.

Remark 1.1.4. The two *elliptic fixed points* in the standard fundamental domain (see §2) are i, fixed by S with stabilizer $\langle S\rangle \cong \mathbb{Z}/2$, and $\rho = e^{2\pi i/3}$, fixed by ST with stabilizer $\langle ST\rangle \cong \mathbb{Z}/3$. Every other point has trivial stabilizer in $\mathrm{PSL}_2(\mathbb{Z})$.

1.2. **Lattices, elliptic curves, and the j-invariant.**

The modular group acts on $\mathscr{H}$ precisely because $\mathscr{H}$ parametrizes lattices up to homothety, hence elliptic curves up to isomorphism over $\mathbb{C}$. The construction proceeds as follows.

Definition 1.2.1. A *lattice* in $\mathbb{C}$ is a subgroup $\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ of rank 2 with $\omega_1, \omega_2 \in \mathbb{C}$ linearly independent over $\mathbb{R}$. We always normalize so that $\tau = \omega_1/\omega_2 \in \mathscr{H}$. A *homothety* is a scaling $\Lambda \mapsto c\Lambda$ for $c \in \mathbb{C}^\times$ such that $J(c\Lambda) \leftrightarrow J(\Lambda)$.

Definition 1.2.2. For a lattice Λ, the *Eisenstein series* and *modular discriminant* are

$$g_2(\Lambda) = 60 \sum_{\substack{\omega\in\Lambda \\ \omega\neq 0}} \omega^{-4}, \qquad g_3(\Lambda) = 140 \sum_{\substack{\omega\in\Lambda \\ \omega\neq 0}} \omega^{-6}, \qquad \Delta(\Lambda) = g_2(\Lambda)^3 - 27\, g_3(\Lambda)^2.$$

One has $\Delta(\Lambda) \neq 0$ for every lattice, and $g_k(c\Lambda) = c^{-2k} g_k(\Lambda)$, $\Delta(c\Lambda) = c^{-12}\Delta(\Lambda)$.

Definition 1.2.3. The *Weierstrass $\wp$-function* of Λ is the meromorphic function

$$\wp(z;\Lambda) = \frac{1}{z^2} + \sum_{\substack{\omega\in\Lambda \\ \omega\neq 0}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right),$$

which is Λ-periodic and satisfies the differential equation $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. The map $z \mapsto (\wp(z), \wp'(z))$ gives an analytic isomorphism

$$\mathbb{C}/\Lambda \xrightarrow{\sim} E_\Lambda \subset \mathbb{P}^2, \qquad E_\Lambda \colon y^2 = 4x^3 - g_2(\Lambda)x - g_3(\Lambda),$$

identifying $\mathbb{C}/\Lambda$ with a smooth projective plane cubic (an elliptic curve over $\mathbb{C}$).

Theorem 1.2.4. *The formula $j(\Lambda) = 1728\, g_2(\Lambda)^3/\Delta(\Lambda)$ defines a holomorphic function $j\colon \mathscr{H} \to \mathbb{C}$ satisfying $j(c\Lambda) = j(\Lambda)$ for all $c \in \mathbb{C}^\times$ and $j(\gamma\cdot\tau) = j(\tau)$ for all $\gamma \in \Gamma(1)$. The induced map*

$$j\colon \Gamma(1)\backslash\mathscr{H} \xrightarrow{\sim} \mathbb{A}^1_{\mathbb{C}}$$

is a complex-analytic isomorphism, uniquely characterized by $j(i) = 1728$, $j(e^{2\pi i/3}) = 0$, and $j(\infty) = \infty$. In particular, two elliptic curves E_Λ and $E_{\Lambda'}$ over $\mathbb{C}$ are isomorphic if and only if $j(\Lambda) = j(\Lambda')$, so j is a complete invariant for the isomorphism class of a complex elliptic curve.

Definition 1.2.5. Setting $q = e^{2\pi i\tau}$ as the local parameter at ∞, the j-function has the q-expansion

$$j(\tau) = q^{-1} + 744 + 196884\,q + 21493760\,q^2 + \cdots \in \mathbb{Z}[[q^{-1}, q]].$$

Compactifying $\Gamma(1)\backslash\mathscr{H}$ by adjoining the single point ∞ yields the compact Riemann surface $\mathbb{P}^1_{\mathbb{C}}$, and j extends to $j\colon \Gamma(1)\backslash\mathscr{H}^* \xrightarrow{\sim} \mathbb{P}^1_{\mathbb{C}}$.

1.3. **The Picard group and the Jacobian.**

Definition 1.3.1. Let X be a compact Riemann surface. We denote the free abelian group (X) to be the group of divisors $D_i \subset X$. The *divisor group* $\mathrm{Div}(X)$ is the free abelian group on the points of X, written $D = \sum_{P\in X} n_P \cdot (P)$ with $n_P \in \mathbb{Z}$ almost all zero. The *degree* is $\deg D = \sum n_P$. The *degree-zero subgroup* is $\mathrm{Div}^0(X) = \ker(\deg)$.

A divisor $(f) = \sum_P \mathrm{ord}_P(f)\cdot(P)$ arising from a nonzero meromorphic function $f \in \mathscr{M}(X)^\times$ is *principal*; the subgroup of principal divisors is $\mathrm{Princ}(X) \subset \mathrm{Div}^0(X)$. The *Picard group* is

$$\mathrm{Pic}(X) = \mathrm{Div}^0(X)/\mathrm{Princ}(X).$$

Theorem 1.3.2 (Abel–Jacobi)**.** *Let X be a compact Riemann surface of genus g. Fix a basis $\omega_1,\dots,\omega_g$ of holomorphic differentials $H^0(X,\Omega^1)$ and a basis $\gamma_1,\dots,\gamma_{2g}$ of $H_1(X,\mathbb{Z})$. The* period lattice *is*

$$\Lambda_X = \left\{ \left(\int_\gamma \omega_1, \dots, \int_\gamma \omega_g \right) \,\middle|\, \gamma \in H_1(X,\mathbb{Z}) \right\} \subset \mathbb{C}^g,$$

a lattice of full rank. The Abel–Jacobi map

$$\mathrm{AJ}\colon\ \mathrm{Div}^0(X) \to \mathbb{C}^g/\Lambda_X, \qquad \sum_i n_i(P_i) \mapsto \sum_i n_i \int_{P_0}^{P_i} (\omega_1,\dots,\omega_g),$$

where P_0 is any base point, descends to an isomorphism of complex Lie groups

$$\mathrm{Pic}(X) \xrightarrow{\sim} J(X) := \mathbb{C}^g/\Lambda_X.$$

The complex torus $J(X)$ is the Jacobian variety *of X; via the Riemann bilinear relations it admits a canonical projective embedding over $\mathbb{C}$.*

Theorem 1.3.3. *Let X be a smooth projective curve defined over a number field k, with a rational base point $O \in X(k)$. Then $J(X)$ has a projective model defined over k, the closed embedding*

$$\iota\colon X \hookrightarrow J(X), \qquad P \mapsto \mathrm{cl}[(P)-(O)],$$

is defined over k, and its image generates $J(X)$ as an abelian variety. The N-torsion satisfies $J(X)[N] \cong (\mathbb{Z}/N\mathbb{Z})^{2g}$, and $k(J(X)[N])/k$ is a finite Galois extension fitting into the commutative diagram

$$\begin{tikzcd}
\mathrm{Gal}(k(J(X)[N])/k) \arrow[r, hook] \arrow[dr, "\rho_N"'] & \mathrm{Aut}(J(X)[N]) \arrow[d, "\sim"] \\
& \mathrm{GL}_{2g}(\mathbb{Z}/N\mathbb{Z})
\end{tikzcd}$$

where ρ_N is the N-adic Galois representation.

1.4. **Divisors with support in a set of cusps.**

Definition 1.4.1. Let $S = \{s_1,\dots,s_r\} \subset X$ be a finite nonempty subset. The *S-divisor group* is the free abelian group $\mathrm{Div}^S(X) = \bigoplus_{i=1}^r \mathbb{Z}\cdot(s_i)$, with degree-zero subgroup $\mathrm{Div}_0^S(X)$. The *cuspidal divisor class group* with support S is

$$\mathrm{Pic}^S(X) = \mathrm{Div}_0^S(X)\big/\left(\mathrm{Div}_0^S(X)\cap\mathrm{Princ}(X)\right),$$

which embeds naturally into $\mathrm{Pic}(X)$, or equivalently into $J(X)$ via Abel–Jacobi. The elements of $\mathrm{Pic}^S(X)$ measure exactly those linear equivalences among the points of S that are witnessed by global meromorphic functions.

Theorem 1.4.2 (Manin–Drinfel'd, [8])**.** *Let X be a modular curve and $S = X^\infty$ its set of cusps. Then* $\mathrm{Pic}^S(X)$ *is a finite group; equivalently, every degree-zero divisor supported on cusps maps to a torsion class in $J(X)$ under the Abel–Jacobi map. The diagram*

$$\begin{tikzcd}
\mathrm{Div}_0^S(X) \arrow[r, "\mathrm{AJ}"] \arrow[d, two heads] & J(X) \\
\mathrm{Pic}^S(X) \arrow[ur, hook, "\textit{finite image}"'] &
\end{tikzcd}$$

commutes and the image of $\mathrm{Pic}^S(X)$ *in $J(X)$ consists entirely of torsion points.*

Remark 1.4.3. When $X = X_1(p)$ and S is the specific set of rational cusps above Q_0 (to be defined in §2), the group $\mathrm{Pic}^S(X_1(p))$ is denoted $\mathscr{C}_1^S(p)$ and is the central object of the Kubert–Lang theorem [7].

1.5. **Characters and Bernoulli numbers.**

Definition 1.5.1. A *Dirichlet character modulo N* is a group homomorphism $\chi\colon (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$, extended to $\mathbb{Z}$ by $\chi(n) = \chi(n \bmod N)$ when $(n,N) = 1$ and $\chi(n) = 0$ otherwise. A character is *even* when $\chi(-1) = 1$, and *odd* when $\chi(-1) = -1$. For an odd prime p, the *Teichmüller character* is the unique $\omega\colon (\mathbb{Z}/p\mathbb{Z})^\times \to \mathbb{Z}_p^\times$ with $\omega(a) \equiv a \pmod{p}$.

Definition 1.5.2. The *Bernoulli polynomials* $\mathbf{B}_k(X) \in \mathbb{Q}[X]$ are defined by the exponential generating series

$$\frac{t\,e^{tX}}{e^t-1} = \sum_{k=0}^{\infty} \mathbf{B}_k(X)\frac{t^k}{k!}.$$

Explicitly, $\mathbf{B}_0 = 1$, $\mathbf{B}_1(X) = X - \frac{1}{2}$, $\mathbf{B}_2(X) = X^2 - X + \frac{1}{6}$. The distribution relation

$$\mathbf{B}_k(X) = N^{k-1}\sum_{j=0}^{N-1} \mathbf{B}_k\left(\frac{X+j}{N}\right)$$

holds for all $N \geq 1$, and is the algebraic reason Bernoulli polynomials appear in p-adic L-functions at every level of the cyclotomic tower.

Fact 1.5.3. *The symmetry $\mathbf{B}_2(1-X) = \mathbf{B}_2(X)$ holds, since*

$$(1-X)^2 - (1-X) + \tfrac{1}{6} = X^2 - X + \tfrac{1}{6}.$$

This forces the second Stickelberger element $\theta^{(2)}$ to be well-defined on $G/\{\pm 1\}$ rather than merely on G.

Definition 1.5.4. For $x \in \mathbb{R}$, write $\langle x \rangle \in [0,1)$ for the fractional part of x. For a Dirichlet character χ modulo N, the *generalized Bernoulli number* attached to χ and weight k is

$$B_{k,\chi} = N^{k-1}\sum_{a=1}^{N} \mathbf{B}_k\big(\langle \tfrac{a}{N} \rangle\big)\,\chi(a).$$

The numbers $B_{2,\chi}$ for even χ of $(\mathbb{Z}/p\mathbb{Z})^\times$ control the orders of the eigenspaces of $\mathscr{C}_1^S(p)$, and their p-adic valuations $\operatorname{ord}_p(B_{2,\bar{\chi}})$ give the exponents in the class number formula of Kubert–Lang [2, §3].

2. Modular Curves

Armed with the foundational language of lattices, elliptic curves, and the Picard group, we now turn to the geometric objects at the heart of the project. Modular curves arise as quotients of $\mathscr{H}$ by congruence subgroups of $\Gamma(1)$, and their remarkable feature is that despite being defined purely analytically, they carry algebraic models over $\mathbb{Q}$ and parametrize elliptic curves with additional arithmetic structure. The theory proceeds in three stages: we describe the fundamental domain of $\Gamma(1)\backslash\mathscr{H}$; introduce the congruence subgroups $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N)$ and their affine modular curves $Y(N)$, $Y_1(N)$, $Y_0(N)$; and compactify these curves by adjoining cusps to obtain the projective modular curves $X(N)$, $X_1(N)$, $X_0(N)$, whose cuspidal divisor class groups will be the central object for the rest of the project. We follow Lang [2, §3] and Shimura [14] throughout.

2.1. **The fundamental domain.**

The quotient $\Gamma(1)\backslash\mathscr{H}$ may be realized concretely by choosing a fundamental domain — a closed region meeting every orbit exactly once in its interior. The standard choice is the region bounded by the vertical lines $\operatorname{Re}(\tau) = \pm\frac{1}{2}$ and the unit semicircle. As Lang observes [2, §3], the coset space $\Gamma(1)\backslash\mathscr{H}$ has this well-known shape, and the j-function gives a complex-analytic isomorphism from it onto the affine line.

Definition 2.1.1. A *fundamental domain* for the action of $\Gamma(1)$ on $\mathscr{H}$ is a closed subset $\mathscr{F} \subset \mathscr{H}$ such that every $\Gamma(1)$-orbit meets $\mathscr{F}$, and two distinct interior points of $\mathscr{F}$ do not lie in the same orbit.

Theorem 2.1.2 ([2, §3])**.** *The region*

$$\mathscr{F} = \left\{\tau \in \mathscr{H} \;\middle|\; |\tau| \geq 1,\ |\operatorname{Re}(\tau)| \leq \tfrac{1}{2}\right\}$$

is a fundamental domain for $\Gamma(1)\backslash\mathscr{H}$. The quotient $\Gamma(1)\backslash\mathscr{H}$ is a non-compact Riemann surface of genus zero.

Remark 2.1.3. The two elliptic fixed points are i, with stabilizer $\langle S \rangle \cong \mathbb{Z}/2$ generated by $\tau \mapsto -1/\tau$, and $\rho = e^{2\pi i/3}$, with stabilizer $\langle ST \rangle \cong \mathbb{Z}/3$ generated by $\tau \mapsto (\tau-1)/\tau$. Every other point of $\mathscr{H}$ has trivial stabilizer in $\mathrm{PSL}_2(\mathbb{Z})$.

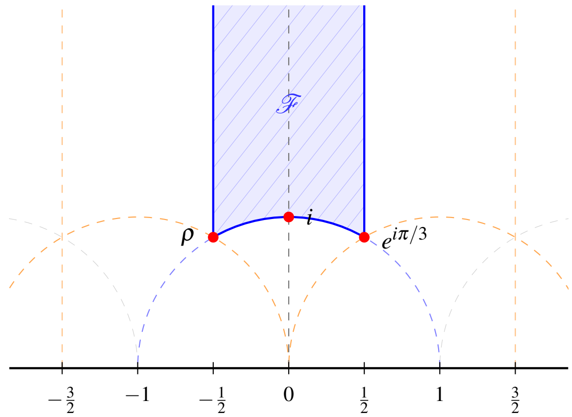


FIGURE 1. The standard fundamental domain $\mathscr{F}$ for $\Gamma(1)\backslash\mathscr{H}$ [2, §3]. The shaded region is $\mathscr{F} = \{\tau \in \mathscr{H} \mid |\tau| \geq 1, |\Re(\tau)| \leq \frac{1}{2}\}$. Solid blue lines are the boundary of $\mathscr{F}$; dashed blue is the full unit circle $|\tau| = 1$; dashed orange are the circles $|\tau \pm 1| = 1$ centered at ± 1 and the vertical walls of the adjacent translates $T^{\pm 1}\mathscr{F}$; dashed gray are the circles $|\tau \pm 2| = 1$. Red dots mark the elliptic fixed points $\rho = e^{2\pi i/3}$, i, and $e^{i\pi/3}$, at which the stabilizers in $\mathrm{PSL}_2(\mathbb{Z})$ are $\langle ST\rangle \cong \mathbb{Z}/3$, $\langle S\rangle \cong \mathbb{Z}/2$, and $\langle TS\rangle \cong \mathbb{Z}/3$ respectively [14].

2.2. **Congruence subgroups.** For each positive integer N — the *level* — one singles out three natural subgroups of $\Gamma(1)$ by imposing congruence conditions modulo N on the matrix entries [2, §3]. The resulting tower $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \Gamma(1)$ induces a tower of coverings of modular curves. The successive quotients

$$\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$$

are explicit finite abelian groups whose characters will later index the eigenspace decomposition of the cuspidal group $\mathscr{C}_1^S(p)$.

Definition 2.2.1 ([2, §3])**.** For $N \geq 1$ and $\det(\gamma) = 1$, define

$$\Gamma(N) = \mathrm{SL}_2(\mathbb{Z}) = \left\{ \gamma \in \Gamma(1) \;\middle|\; \gamma \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \pmod N,\ \det(\gamma) = 1 \right\},$$

$$\Gamma_1(N) = \left\{ \gamma \in \Gamma(1) \;\middle|\; \gamma \equiv \begin{bmatrix} 1 & * \\ 0 & 1 \end{bmatrix} \pmod N \right\}, \qquad \Gamma_0(N) = \left\{ \gamma \in \Gamma(1) \;\middle|\; \gamma \equiv \begin{bmatrix} * & * \\ 0 & * \end{bmatrix} \pmod N \right\}.$$

Proposition 2.2.2 ([2, §3])**.** *The inclusions $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \Gamma(1)$ fit into short exact sequences*

$$1 \to \Gamma(N) \to \Gamma(1) \xrightarrow{\bmod N} \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \to 1,$$

$$1 \to \Gamma_1(N) \to \Gamma_0(N) \xrightarrow{a \bmod N} (\mathbb{Z}/N\mathbb{Z})^\times \to 1,$$

giving in particular $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$. The tower of inclusions fits into the commutative diagram

$$\begin{tikzcd} \Gamma(N) \arrow[r, hook] \arrow[d, hook] & \Gamma_1(N) \arrow[r, hook] \arrow[d, hook] & \Gamma_0(N) \arrow[r, hook] \arrow[d, hook] & \Gamma(1) \arrow[d, hook] \\ \mathrm{SL}_2(\mathbb{Z}) \arrow[r, equal] & \mathrm{SL}_2(\mathbb{Z}) \arrow[r, equal] & \mathrm{SL}_2(\mathbb{Z}) \arrow[r, equal] & \mathrm{SL}_2(\mathbb{Z}) \end{tikzcd}$$

in which the bottom row is the identity and the vertical arrows are all inclusions.

Remark 2.2.3. The $\mathrm{SL}_2(\mathbb{Z})$ condition $\det(\gamma) = 1$ forces $d \equiv a^{-1} \pmod N$, so the reduction map $\gamma \mapsto a \bmod N$ on $\Gamma_0(N)$ lands in the units $(\mathbb{Z}/N\mathbb{Z})^\times$ rather than all of $\mathbb{Z}/N\mathbb{Z}$.

2.3. **Affine modular curves.** Each congruence subgroup Γ gives rise to a Riemann surface $\Gamma\backslash\mathscr{H}$ which, as Lang explains [2, §3], can be projectively embedded as an affine algebraic curve via a suitable extension of the j-invariant. When Γ is one of the three standard subgroups, the resulting curves are denoted $Y(N)$, $Y_1(N)$, $Y_0(N)$; the subscript $\mathbb{C}$ will indicate the Riemann surface of complex points.

Definition 2.3.1 ([2, §3])**.** For a congruence subgroup $\Gamma \subseteq \Gamma(1)$, the *affine modular curve*

$$Y(\Gamma) = \Gamma\backslash\mathscr{H}$$

is the Riemann surface of Γ-orbits in $\mathscr{H}$. The coordinate functions of a projective embedding $f_\Gamma \colon \Gamma\backslash\mathscr{H} \hookrightarrow \mathbb{P}^n$ can be chosen in many ways suited to different applications; the image is always an affine algebraic curve. For the three standard subgroups, write

$$Y(N) = \Gamma(N)\backslash\mathscr{H}, \qquad Y_1(N) = \Gamma_1(N)\backslash\mathscr{H}, \qquad Y_0(N) = \Gamma_0(N)\backslash\mathscr{H}.$$

The inclusions of congruence subgroups induce a tower of finite coverings, summarized in the commutative diagram

$$\begin{tikzcd}
Y(N)_{\mathbb{C}} \arrow[r, "\sim"] \arrow[d] & \Gamma(N)\backslash\mathscr{H} \arrow[d] \\
Y_1(N)_{\mathbb{C}} \arrow[r, "\sim"] \arrow[d] & \Gamma_1(N)\backslash\mathscr{H} \arrow[d] \\
Y_0(N)_{\mathbb{C}} \arrow[r, "\sim"] \arrow[d] & \Gamma_0(N)\backslash\mathscr{H} \arrow[d] \\
Y(1)_{\mathbb{C}} \arrow[r, "\sim"] & \Gamma(1)\backslash\mathscr{H}
\end{tikzcd}$$

where each vertical arrow is a finite possibly-ramified covering, the degree of $Y_1(N) \to Y_0(N)$ being $[(\mathbb{Z}/N\mathbb{Z})^\times : \{\pm 1\}]$.

2.4. **Moduli interpretation.** The affine modular curves acquire arithmetic meaning through their interpretation as moduli spaces [2, §3]. Points of $Y(1)$ correspond to isomorphism classes of elliptic curves over $\mathbb{C}$; the finer objects $Y_0(N)$ and $Y_1(N)$ parametrize elliptic curves with level-N structure. This interpretation is what permits the descent from Riemann surfaces over $\mathbb{C}$ to smooth affine curves over $\mathbb{Q}$, first made precise by Shimura [14] and given integral models by Deligne–Rapoport [9] and Katz–Mazur [10].

Theorem 2.4.1 ([2, §3],[14])**.** *The complex points admit the following moduli interpretations.*

(1) $Y(1)_{\mathbb{C}}$ *parametrizes isomorphism classes of elliptic curves A over $\mathbb{C}$, via $A \mapsto j(A)$.*
(2) $Y_0(N)_{\mathbb{C}}$ *parametrizes isomorphism classes of pairs (A,Z), where A is an elliptic curve over $\mathbb{C}$ and $Z \subset A$ is a cyclic subgroup of order N.*
(3) $Y_1(N)_{\mathbb{C}}$ *parametrizes isomorphism classes of pairs (A,P), where A is an elliptic curve over $\mathbb{C}$ and $P \in A$ has exact order N.*
(4) $Y(N)_{\mathbb{C}}$ *parametrizes isomorphism classes of triples $(A,(P,Q))$, where (P,Q) is an ordered $\mathbb{Z}/N\mathbb{Z}$-basis of the N-torsion $A[N]$.*

Analytically, for $A = \mathbb{C}/[\tau,1]$, one takes $P = [1/N]$ and $Z = \langle P\rangle$. The moduli interpretations are compatible with the covering maps via

$$\begin{tikzcd}
Y_1(N) \arrow[r, "(A,P)\mapsto(A,\langle P\rangle)"] \arrow[d] & Y_0(N) \arrow[d, "(A,Z)\mapsto A"] \\
Y(1) \arrow[r, equal] & Y(1)
\end{tikzcd}$$

Theorem 2.4.2 ([9, 10])**.** *The modular curves $Y(N)$, $Y_1(N)$, $Y_0(N)$ descend to smooth affine curves defined over $\mathbb{Q}$, with integral models over $\mathbb{Z}[1/N]$. The covering maps and the moduli interpretations are all defined over $\mathbb{Q}$.*

2.5. **Compactification and cusps.** The affine modular curves are non-compact, missing finitely many points corresponding to degenerate elliptic curves. These missing points — the *cusps* — are encoded analytically by Γ-orbits on $\mathbb{Q} \cup \{\infty\}$ [2, §3]. As described there, a typical neighborhood of ∞ is the part of the upper half-plane lying above a horizontal line, and a typical neighborhood of a rational cusp r is a disc in $\mathscr{H}$ tangent to the real axis at r.

Definition 2.5.1. The *extended upper half-plane* is

$$\mathscr{H}^* = \mathscr{H} \cup \mathbb{Q} \cup \{\infty\},$$

with the topology in which neighborhoods of ∞ are of the form $\{\tau \in \mathscr{H} \mid \operatorname{Im}(\tau) > M\} \cup \{\infty\}$ for $M > 0$, and neighborhoods of $r \in \mathbb{Q}$ are open discs in $\mathscr{H}$ tangent to $\mathbb{R}$ at r together with $\{r\}$.

FIGURE 2. Typical neighborhoods of the two types of cusps in $\mathscr{H}^*$ [2, §3]. *Left*: a neighborhood of ∞ is the region $\{\tau \in \mathscr{H} \mid \operatorname{Im}(\tau) > M\}$ above a horizontal line. *Right*: a neighborhood of $r \in \mathbb{Q}$ is an open disc of radius ε in $\mathscr{H}$ tangent to $\mathbb{R}$ at r, together with $\{r\}$ [14].

Theorem 2.5.2 ([2, §3])**.** *For every congruence subgroup $\Gamma \subseteq \Gamma(1)$, the quotient $\Gamma \backslash \mathscr{H}^*$ is a compact Riemann surface. The* projective modular curve *is*

$$X(\Gamma) = Y(\Gamma) \cup X^\infty(\Gamma), \qquad X^\infty(\Gamma) = \Gamma \backslash (\mathbb{Q} \cup \{\infty\}),$$

where $X^\infty(\Gamma)$ is the finite set of cusps. *For the standard subgroups, write $X(N)$, $X_1(N)$, $X_0(N)$, and note that each ramified covering $Y_1(N) \to Y_0(N)$ extends to a ramified covering $X_1(N) \to X_0(N)$ of compact Riemann surfaces.*

Theorem 2.5.3 ([2, §3])**.** *Under the identification $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$, the covering $X_1(N) \to X_0(N)$ is Galois with covering group $G/\{\pm I\}$, where $G = (\mathbb{Z}/N\mathbb{Z})^\times$ acts by*

$$a \mapsto \gamma_a, \qquad \gamma_a \equiv \begin{bmatrix} a & 0 \\ 0 & a^{-1} \end{bmatrix} \pmod{N},$$

and γ_a descends to $(A, P) \mapsto (A, aP)$ on $Y_1(N)$. The covering fits into the diagram

$$\begin{tikzcd} X_1(N) \arrow[r, hook] \arrow[d, "\text{Galois, group } G/\{\pm I\}"'] & X_1(N) \\ X_0(N) \arrow[r, hook] & X_0(N) \end{tikzcd}$$

2.6. **Cusps of $X_0(p)$ and $X_1(p)$ for prime level.** For the remainder of the project we fix a prime $p \geq 3$. The structure of the cusps at prime level is particularly clean, and Lang [2, §3] notes that there are precisely two cusps on $X_0(p)$ lying above $j = \infty$.

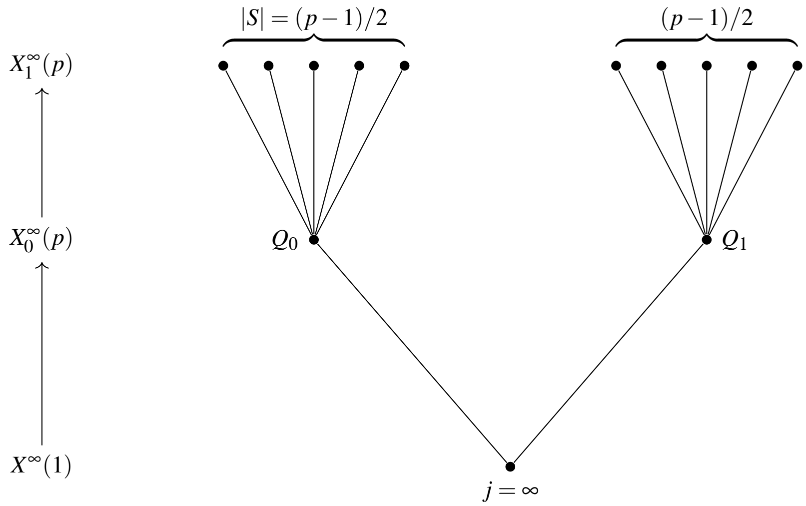
Theorem 2.6.1 ([2, §3])**.** *The curve $X_0(p)$ has exactly two cusps above $j = \infty$, namely*

$$Q_0 = [0], \qquad Q_1 = [\infty],$$

the $\Gamma_0(p)$-orbits of 0 and ∞ in $\mathbb{Q} \cup \{\infty\}$. The covering degree and cusp structure of $X_1(p) \to X_0(p)$ are governed by

$$[X_1(p) : X_0(p)] = \frac{p-1}{2}.$$

This covering is unramified over Q_0 and Q_1, so there are exactly $(p-1)/2$ cusps of $X_1(p)$ above each of Q_0 and Q_1, for a total of $p-1$ cusps on $X_1(p)$.



Definition 2.6.2 ([2, §3])**.** Let $S \subset X_1^\infty(p)$ denote the set of $(p-1)/2$ cusps lying above Q_0,

$$S = \{x \in X_1^\infty(p) \mid x \mapsto Q_0 \text{ under } X_1(p) \to X_0(p)\}.$$

The cusps $Q_0 \in X_0(p)$ and the elements of $S \subset X_1(p)$ are the *rational cusps*, meaning they are defined over $\mathbb{Q}$ in the moduli-theoretic model.

2.7. **The cuspidal divisor class group.** With the rational cusps $S \subset X_1^\infty(p)$ identified, we can now define the central object of the project. The cuspidal divisor class group $\mathscr{C}_1^S(p)$ sits inside the Jacobian $J_1(p) = \mathrm{Jac}(X_1(p))$ as a finite torsion subgroup by the Manin–Drinfel'd theorem, and its structure is the subject of the Kubert–Lang theorem in the next section.

Definition 2.7.1. The *cuspidal divisor class group* relative to S is

$$\mathscr{C}_1^S(p) = \mathrm{Pic}^S(X_1(p)) = \frac{\mathrm{Div}_0^S(X_1(p))}{\mathrm{Div}_0^S(X_1(p)) \cap \mathrm{Princ}(X_1(p))},$$

sitting inside the Jacobian $J_1(p) = \mathrm{Jac}(X_1(p))$ via the Abel–Jacobi map. By Theorem 1.4.2, this group is finite. The full cuspidal group $\mathscr{C}^\infty(p) = \mathrm{Pic}^\infty(X_1(p))$ uses all $p-1$ cusps.

Remark 2.7.2. The key structural fact, proved by Lang [2, §3] following Kubert–Lang [7], is that $\mathscr{C}_1^S(p)$ is not merely abstractly finite — its structure is completely determined by second-order Bernoulli numbers $B_{2,\chi}$, and it is isomorphic as a $\mathbb{Z}[G/\{\pm 1\}]$-module to the co-Stickelberger module $R_0/\mathscr{S}^{(2)}$ to be constructed in §4. The covering maps and cuspidal groups fit into the commutative diagram

$$\begin{tikzcd}
\mathscr{C}_1^S(p) \arrow[r, hook] \arrow[d] & J_1(p) \arrow[d] \\
\mathscr{C}^\infty(p) \arrow[r, hook] & J_1(p)
\end{tikzcd}$$

where the left vertical map is the natural inclusion and the right is the identity.

3. The Kubert–Lang Theory and Mazur–Wiles

The two preceding sections have established the geometric and arithmetic actors separately. The modular curve $X_1(p)$ carries the cuspidal group $\mathscr{C}_1^S(p) \subset J_1(p)$; the cyclotomic field $\mathbb{Q}(\mu_p)$ carries the class group $\mathrm{Cl}(\mathbb{Q}(\mu_p))^-$ and its Stickelberger annihilator. The purpose of this section is to show that these two objects are the same, and to draw the consequence that cuspidal torsion points on $J_1(p)$ generate unramified abelian extensions of cyclotomic fields — the Iwasawa Main Conjecture for $\mathbb{Q}$.

3.1. **The Stickelberger element and the co-Stickelberger module.** Let $p \geq 5$ be prime, $G = (\mathbb{Z}/p\mathbb{Z})^\times$, and $\sigma_a \in \mathrm{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$ the element sending $\zeta_p \mapsto \zeta_p^a$.

Definition 3.1.1 ([2, §3]). The *Stickelberger element of order* 2 is

$$\theta^{(2)} = p \sum_{a \in G/\{\pm 1\}} \tfrac{1}{2}\,\mathbf{B}_2\left(\left\langle \tfrac{a}{p} \right\rangle\right) \sigma_a^{-1} \;\in\; \mathbb{Q}[G/\{\pm 1\}].$$

Setting $R = \mathbb{Z}[G/\{\pm 1\}]$, $R_0 = \ker(\varepsilon\colon R \to \mathbb{Z})$ the augmentation ideal, and $I^{(2)} = \langle \sigma_c - c^2 \mid (c, 6p) = 1 \rangle \subset R$, the *Stickelberger ideal* is

$$\mathscr{S}^{(2)} = R\theta^{(2)} \cap R_0 = (I^{(2)} \cap R_0) \cdot \theta^{(2)},$$

and the *co-Stickelberger module* is $R_0/\mathscr{S}^{(2)}$.

Remark 3.1.2. The symmetry $\mathbf{B}_2(1-x) = \mathbf{B}_2(x)$ is precisely what makes $\theta^{(2)}$ well-defined on $G/\{\pm 1\}$ rather than merely on G. The analogous first-order element $\theta^{(1)}$, built from $\mathbf{B}_1(x) = x - \frac{1}{2}$, governs the minus part of the cyclotomic class group via the classical Stickelberger theorem [2, §2]; the second-order element governs the cuspidal group via Kubert–Lang.

3.2. **The Kubert–Lang theorem.**

Theorem 3.2.1 (Kubert–Lang, [7],[2, §3]). *There is a natural isomorphism of R-modules*

$$R_0/\mathscr{S}^{(2)} \xrightarrow{\sim} \mathscr{C}_1^S(p),$$

and the order of $\mathscr{C}_1^S(p)$ *is given by the class number formula*

$$|\mathscr{C}_1^S(p)| = (R_0 : \mathscr{S}^{(2)}) = p \prod_{\substack{\chi \neq 1 \\ \chi \text{ even}}} \pm \tfrac{1}{2} B_{2,\chi},$$

where χ *ranges over nontrivial even characters of* $(\mathbb{Z}/p\mathbb{Z})^\times$ *and* $B_{2,\chi} = p\sum_{a=1}^{p-1} \mathbf{B}_2(\langle a/p \rangle)\chi(a)$.

Theorem 3.2.2 ([7]). *The p-primary component* $\mathscr{C}_1^S(p)^{(p)}$ *is isomorphic to* $M = \mathbb{Z}_p[G]_0/\mathscr{S}_p^{(2)}$ *as* $\mathbb{Z}_p[G]$*-modules. The eigenspace decomposition under* $G = (\mathbb{Z}/p\mathbb{Z})^\times$ *is*

$$M(\chi) \cong \begin{cases} 0 & \chi = 1 \text{ or } \chi = \omega^2, \\ \mathbb{Z}_p/B_{2,\bar{\chi}}\mathbb{Z}_p & \chi \text{ even},\ \chi \neq 1, \omega^2, \end{cases}$$

so $M(\chi)$ *is cyclic of order* $p^{n(\chi)}$ *where* $n(\chi) = \mathrm{ord}_p(B_{2,\bar{\chi}})$.

3.3. **The Hecke algebra and the cuspidal ideal.**

Definition 3.3.1 ([6],[2, §4]). The *Hecke algebra* is the commutative $\mathbb{Z}_p$-subalgebra

$$\mathbf{T} \subset \mathrm{End}(J_1(p))$$

generated by the Hecke operators T_ℓ for $\ell \nmid p$, the operator U_p, and the diamond operators $\langle a \rangle$ for $a \in (\mathbb{Z}/p\mathbb{Z})^\times$. For an odd character $\chi \neq \omega, \bar{\omega}$ of G, the *cuspidal ideal* $\mathbf{I}_{\chi\omega} \subset \mathbf{T}$ is the annihilator of the cyclic cuspidal subgroup $Z = \mathscr{C}_1^S(p)(\chi\omega)^{(p)} \cong \mathbb{Z}_p/B_{2,\overline{\chi\omega}}\mathbb{Z}_p$. By Theorem of Wiles [4], it is generated by

$$T_\ell - (1 + \ell\langle \ell \rangle), \qquad U_p - 1, \qquad \langle a \rangle - \chi\omega(a), \qquad p^n,$$

where $p^n = \mathrm{ord}_p(B_{2,\overline{\chi\omega}})$ is the exact power of p dividing the relevant Bernoulli number.

3.4. **Mazur–Wiles and the Iwasawa Main Conjecture.** We arrive at the theorem that is the culmination of the entire project. Let χ be an odd character of G with $\chi \neq \omega, \bar{\omega}$, and define $\mathfrak{g} = \ker(\mathbf{I}_{\chi\omega} \curvearrowright J_1(p))$ to be the finite group of $\mathbf{I}_{\chi\omega}$-torsion points in $J_1(p)(\overline{\mathbb{Q}})$.

Theorem 3.4.1 (Mazur–Wiles, [3],[2, §4]). *The field* $\mathbb{Q}(\mu_{p^n})(\mathfrak{g})$ *is an unramified abelian extension of* $\mathbb{Q}(\mu_{p^n})$ *of degree* $p^n = |Z|$.

Theorem 3.4.2 (Iwasawa Main Conjecture, [3]). *Let $\mathbb{Q}_\infty = \bigcup_n \mathbb{Q}(\mu_{p^n})$, $\Gamma_{\mathrm{cyc}} = \mathrm{Gal}(\mathbb{Q}_\infty/\mathbb{Q}(\mu_p)) \cong \mathbb{Z}_p$, and $\Lambda = \mathbb{Z}_p[[\Gamma_{\mathrm{cyc}}]] \cong \mathbb{Z}_p[[T]]$ the Iwasawa algebra. Let $X_\infty^- = \mathrm{Gal}(L_\infty/\mathbb{Q}_\infty)^-$ where L_∞ is the maximal unramified abelian p-extension. Then for every odd character χ of $(\mathbb{Z}/p\mathbb{Z})^\times$,*

$$\mathrm{char}_\Lambda(X_\infty^-(\chi)) = (f_{\chi\omega}(T)) \subset \Lambda,$$

where $f_{\chi\omega}(T) \in \Lambda$ is the Kubota–Leopoldt power series interpolating the values $L_p(1-n, \chi\omega^{1-n}) = -(1-\chi\omega^{1-n}(p)p^{n-1})\frac{B_{n,\chi\omega^{1-n}}}{n}$.

Remark 3.4.3 (Proof strategy and Herbrand–Ribet). Theorem 3.4.1 is proved by constructing, for each level p^ν and each odd $\chi \neq \omega, \bar{\omega}$, unramified extensions of $\mathbb{Q}(\mu_{p^\nu})$ via the Galois representation on $\mathfrak{g}$, which is upper-triangular with diagonal characters 1 and $\chi\omega$. The off-diagonal cocycle defines the required unramified extension. Passing to the projective limit over ν and comparing characteristic polynomials via the Eichler–Shimura relation $\mathrm{Frob}_\ell^2 - T_\ell \mathrm{Frob}_\ell + \ell\langle\ell\rangle = 0$ yields the Main Conjecture.

At the bottom of the tower ($n = 1$), the Main Conjecture recovers Herbrand–Ribet [5]: for $3 \leq k \leq p-2$ odd,

$$p \mid B_k \iff \mathrm{Cl}(\mathbb{Q}(\mu_p))^{(p)}(\omega^{1-k}) \neq 0.$$

The implication $(\Leftarrow)$ is Herbrand's theorem; the converse $(\Rightarrow)$ is Ribet's, proved via Eisenstein series and modular Galois representations, and is the prototype for the entire Mazur–Wiles construction.

Remark 3.4.4 (The cyclotomic–modular dictionary). The arithmetic dictionary underlying the proof may be summarized as

$$\begin{array}{ccc} \text{Cyclotomic} & \longleftrightarrow & \text{Modular} \\ & & \\ E_{\mathrm{cyc}} \subset \mathscr{O}_{\mathbb{Q}(\mu_p)}^\times & \longleftrightarrow & S \subset X_1^\infty(p) \\ \downarrow & & \downarrow \\ \mathrm{Cl}(\mathbb{Q}(\mu_p))^- & \longleftrightarrow & \mathscr{C}_1^S(p) \\ \uparrow & & \uparrow \\ \mathscr{S}^{(1)} \text{ annihilates} & \longleftrightarrow & \mathscr{S}^{(2)} \text{ presents} \end{array}$$

The crucial asymmetry in the bottom row is that the first-order Stickelberger ideal only *annihilates* the cyclotomic class group, while the second-order Stickelberger ideal completely *presents* the cuspidal group. It is this stronger identification — the Kubert–Lang isomorphism $R_0/\mathscr{S}^{(2)} \cong \mathscr{C}_1^S(p)$ — that makes the modular proof of the Main Conjecture possible.

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